MAT/MATH 215, Final, 13.01.2023, 09:30 - 11:30

Name and Surname (İsim ve Soyisim) : _____

ID (Öğrenci No): : _____

Bölüm (Department) : _____

* This is a closed book and closed notes exam (Bu sınavda kitap ve not kullanılamaz).
* No calculators, no talking and no questions (Hesap makinası, konuşmak ve soru sormak yasaktır).

1	2	3	4	5	Toplam
20	20	20	20	20	100

1.(20 p.) Consider the following differential equation:

$$y''(t) + y'(t) = \cos(t) + \int_0^t (t-\varepsilon)y'(\varepsilon)d\varepsilon$$

If $F(s) = L[y]$, calculate $F(1)$ under the initial conditions $y(0) = y'(0) = 0$.

(Yukarıda verilen diferansiyel denklemi göz önünde bulundurun. $F(s) = L[y]$ ise, $y(0) = y'(0) = 0$ başlangıç koşulları altında, $F(1)$'i hesaplayın.)

Solution : Let $I := \int_0^t (t-\varepsilon)y'(\varepsilon)d\varepsilon$. As we know $I = t \star y' = L^{-1}[L[t]L[y']]$, so

$$L[I] = \frac{1}{s^2}(sL[y] - y(0)) = \frac{1}{s}L[y]$$

Now apply Laplace transform to the given equation to observe that:

$$s^2L[y] + sL[y] = \frac{s}{s^2+1} + \frac{1}{s}L[y]$$

Now solve the relation for $L[y]$ to observe that $F(s) = L[y] = \frac{s^2}{(s^2+1)(s^3+s^2-1)}$. Thus $F(1) = \frac{1}{2}$.

2.(20 p.) Find the general solution of the following system of differential equations.

$$\begin{cases} x' = x \\ y' = 5x + y + 2z \\ z' = 2x - 2y + z \end{cases}$$

(Yukarıda verilen diferansiyel denklem sisteminin genel çözümünü bulunuz.)

Solution: Let's find the solutions $x = \alpha e^{\lambda t}$, $y = \beta e^{\lambda t}$ and $z = \gamma e^{\lambda t}$.

$|A - \lambda\, I| = \begin{vmatrix} 1-\lambda & 0 & 0 \\ 5 & 1-\lambda & 2 \\ 2 & -2 & 1-\lambda \end{vmatrix} = 0 \Rightarrow \lambda_1 = 1,\ \lambda_{2,3} = 1 + 2i,\ 1 - 2i$

For $\lambda_1 = 1$, $(A - \lambda\, I) \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}. \Rightarrow \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} \beta \\ \beta \\ -\frac{5}{2}\beta \end{bmatrix} = 2\beta \begin{bmatrix} 2 \\ 2 \\ -5 \end{bmatrix}$, so the solution is $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2e^t \\ 2e^t \\ -5e^t \end{bmatrix}$.

For $\lambda_{2,3} = 1 + 2i,\ 1 - 2i$, $\begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} 0 \\ \beta \\ i\beta \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + i \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

the solutions are $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ e^t \cos 2t \\ -e^t \sin 2t \end{bmatrix}$ and $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ e^t \sin 2t \\ e^t \cos 2t \end{bmatrix}$.

Thus, the general solution is:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = C_1 \begin{bmatrix} 2e^t \\ 2e^t \\ -5e^t \end{bmatrix} + C_2 \begin{bmatrix} 0 \\ e^t \cos 2t \\ -e^t \sin 2t \end{bmatrix} + C_3 \begin{bmatrix} 0 \\ e^t \sin 2t \\ e^t \cos 2t \end{bmatrix} = \begin{bmatrix} 2C_1 e^t \\ 2C_1 e^t + C_2 e^t \cos 2t + C_3 e^t \sin 2t \\ -5C_1 e^t - C_2 e^t \sin 2t + C_3 e^t \cos 2t \end{bmatrix}$$

3.(20 p.) Consider the following non-homogeneous Cauchy-Euler equation:

$$x^2y'' - 2xy' + 2y = 3x^2, \quad x > 0$$

a) Find the general solution of the associated homogeneous equation.
b) Find a particular solution for the given differential equation by using the method of undetermined coefficients.

Solution: a)Let $x = e^t$, then $t = \ln x$,

$$x^2y'' = \frac{d^2y}{dt^2} - \frac{dy}{dt} \text{ and } xy' = \frac{dy}{dt}.$$

The differential equation transform into

$$\frac{d^2y}{dt^2} - 3\frac{dy}{dt} + 2y = 0.$$

The characteristic equation of this

$$r^2 - 3r + 2 = 0 \Rightarrow r_1 = 1, r_2 = 2 \Rightarrow x_1(t) = e^t \text{ and } x_2(t) = e^{2t}.$$

Since $t = \ln x$, we obtain $x_1(x) = x$ and $x_2(x) = x^2$. Then, the general solution of the associated homogeneous equation is

$$y(x) = c_1x + c_2x^2$$

Second Method: Let $y = x^r$ then $y' = rx^{r-1}$ and $y'' = r(r-1)x^{r-2}$. Subtituting these values in equation we obtain

$$x^2y'' - 2xy' + 2y = r(r-1)x^r - 2rx^r + 2x^r = 0 \Rightarrow x^r(r(r-1) - 2r + 2) = 0 \Rightarrow r^2 - 3r + 2 = 0$$

Then, $r_1 = 1, r_2 = 2$ and the solutions are $y = x^1$ and $y = x^2$. Then, the general solution of the associated homogeneous differential equation is

$$y(x) = c_1x + c_2x^2$$

b) Let Let $x = e^t$, then $t = \ln x$, and the differential equation transform

$$\frac{d^2y}{dt^2} - 3\frac{dy}{dt} + 2y = 3e^{2t}.$$

To calculate particular solution of this equation, we can use the method of undetermined coefficients. $y_p(t) = Ate^{2t}$ must have this form. Substituting this solution above, we get

$$4Ae^{2t} + 4Ate^{2t} - 3(Ae^{2t} + 2Ate^{2t}) + 2Ate^{2t} = 3e^{2t} \Rightarrow A = 3.$$

Then

$$y_p(t) = 3te^{2t} \Rightarrow y_p(x) = 3x^2 \ln x.$$

4.(20 p.) Using reduction of order method find the general solution to $x\dfrac{d^2y}{dx^2}-(x+2)\dfrac{dy}{dx}+2y=0$, given that $y=e^x$ is a solution.

(Çözümü $y=e^x$ olarak verilen $x\dfrac{d^2y}{dx^2}-(x+2)\dfrac{dy}{dx}+2y=0$ denkleminin genel çözümünü Mertebe indirgeme yöntemini kullanarak bulunuz.)

Sol. The solution of in the form $y=c(x)e^x$, say $c(x)=c$

$\dfrac{dy}{dx}=e^xc+e^xc'$ and $\dfrac{d^2y}{dx^2}=e^xc+2e^xc'+e^xc''\Rightarrow$

$$\begin{aligned} x\left(e^xc+2e^xc'+e^xc''\right)-(x+2)\left(e^xc+e^xc'\right)+2ce^x &= 0\\ xe^xc''+(x-2)c'e^x &= 0\\ c''+\frac{(x-2)}{x}c'e^x &= 0 \end{aligned}$$

first order linear equation in $c'(x)$

Integrating multiple $e^{\int\frac{(x-2)}{x}dx}=e^{-e^{\int\left(1-\frac{2}{x}\right)dx}}=e^{x-2\ln x}=x^{-2}e^x$, Multiply linear D.E. with this intergrating multiple, then we have

$$\begin{aligned} x^{-2}e^xc''+x^{-2}e^x\frac{(x-2)}{x}c'e^x &= 0\\ \left(x^{-2}e^xc'\right)' &= 0\\ x^{-2}e^xc' &= k\\ c &= k\int x^2e^{-x}dx+m\\ c &= -ke^{-x}\left(x^2+2x+2\right)+m\\ y &= c(x)e^x=-k\left(x^2+2x+2\right)+me^x \end{aligned}$$

5.(20 p.)
Find all singular points of the following differential equations and classify them as regular singular points or irregular singular points:

$$(1-x^2)y'' - xy' + y = 0$$

Find (at least the first 5 terms of) the series solution about $x = 0$.

Yukarıdaki diferansiyel denklemin bütün tekil noktalarını bulunuz ve regüler yada regüler olmayan olarak sınıflandırınız. Denklemin $x = 0$ noktasındaki seri çözümünü (en az ilk 5 terimi içerek şekilde) bulunuz.

Sol. The standard form of equation is

$$y'' - \frac{x}{1-x^2}y' + \frac{1}{1-x^2}y = 0.$$

Since $p(x) = \frac{x}{1-x^2}$ and $q(x) = \frac{1}{1-x^2}$, it follows that $x = 1$ and $x = -1$ are the singular points.

The point $x = 1$ is a regular singular point since

$$\lim_{x\to 1} \frac{(x-1)x}{1-x^2} = \frac{1}{2} \qquad \lim_{x\to 1} \frac{(x-1)^2}{1-x^2} = 0.$$

The point $x = -1$ is a regular singular point since

$$\lim_{x\to -1} \frac{(x-1)x}{1-x^2} = \frac{1}{2} \qquad \lim_{x\to -1} \frac{(x-1)^2}{1-x^2} = 0.$$

Since $x = 0$ is an ordinary point,

$$y = \sum_{n=0}^{\infty} a_n x^n \qquad y' = \sum_{n=1}^{\infty} a_n n x^{n-1} \qquad y'' = \sum_{n=2}^{\infty} a_n n(n-1)x^{n-2}.$$

Now, we plug into the original equation

$$(1-x^2)\sum_{n=2}^{\infty} a_n n(n-1)x^{n-2} - x\sum_{n=1}^{\infty} a_n n x^{n-1} + \sum_{n=0}^{\infty} a_n x^n = 0$$

$$\sum_{n=2}^{\infty} a_n n(n-1)x^{n-2} - \sum_{n=2}^{\infty} a_n n(n-1)x^n - \sum_{n=1}^{\infty} a_n n x^n \sum_{n=0}^{\infty} a_n x^n = 0$$

$$\sum_{n=0}^{\infty} a_{n+2} n + 2(n+1)x^n - \sum_{n=2}^{\infty} a_n n(n-1)x^n - \sum_{n=1}^{\infty} a_n n x^n \sum_{n=0}^{\infty} a_n x^n = 0$$

$$[2a_2 + 6a_3 x - a_1 x + a_0 + a_1 x] + \sum_{n=2}^{\infty} [a_{n+2}(n+2)(n+1) - a_n n(n-1) - a_n n + a_n]x^n = 0$$

Then, we have

$$[(2a_2 + a_0) + (6a_3)x] = 0 \qquad \sum_{n=2}^{\infty} [a_{n+2}(n+2)(n+1) + a_n(1-n^2)]x^n = 0$$

Hence, we get

$$2a_2 + a_0 = 6a_3 = a_{n+2}(n+2)(n+1) + a_n(1-n^2) = 0.$$

It follows that $a_3 = 0$, $a_0 = -2a_2$ and the recurrence relation

$$a_{n+2} = \frac{n^2 - 1}{(n+1)(n+2)} a_n = \frac{n-1}{n+2} a_n, \qquad \text{for } n = 2, 3, ...$$

Note here that from the recurrence relation we get $0 = a_3 = a_5 = a_7 = ...$
Furthermore,

$$a_2 = \frac{-1}{2} a_0, \qquad a_4 = \frac{1}{4} a_2 = \frac{-1}{8} a_0, \qquad a_6 = \frac{1}{2} a_4 = \frac{-1}{16} a_0.$$

Therefore, the general solution is

$$y = a_0 + a_1 x - \frac{1}{2} a_0 x^2 - \frac{1}{8} a_0 x^4 - \frac{1}{6} a_0 x^6 .....$$